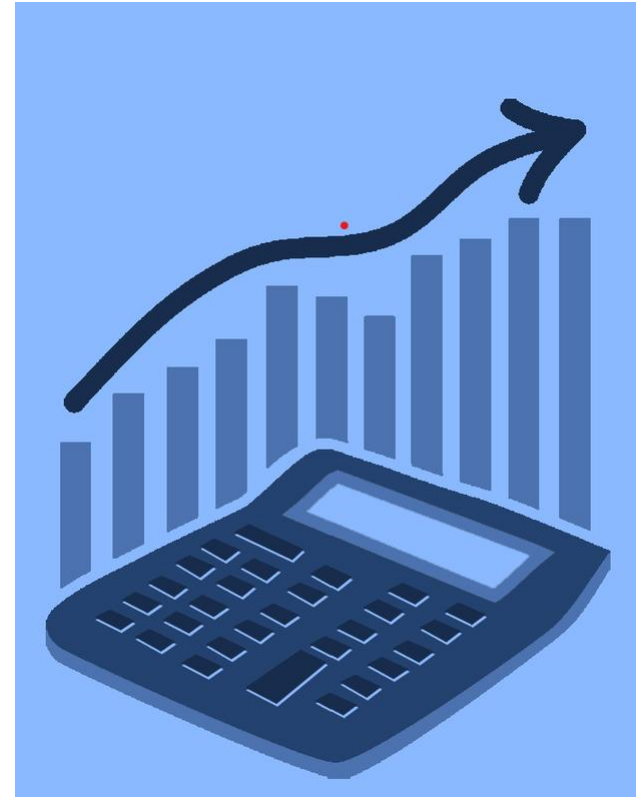


Corporate Valuation Methods

Session 1

Valuation Foundations & Preparing for Valuation

From reported financial statements to valuation-relevant operating economics and cash flows



What You Should Be Able to Do After Session 1

DISTINGUISH

Price, enterprise value and equity value

NORMALIZE

Reported earnings into sustainable operating earnings

MEASURE

NOPAT, invested capital and ROIC

CONNECT

Growth to reinvestment and ROIC

CONVERT

Operating earnings into FCFF and FCFE

PREPARE

The operating inputs required for forecasting and valuation.

Before we discount cash flows, we need to know whether the cash flows themselves make economic sense.

Price Is Observable. Value Is Estimated.

MARKET PRICE

The price at which the security trades today. Observable, but not necessarily equal to intrinsic value.

ESTIMATED VALUE

An analyst's estimate based on future economics, risk and assumptions.

TRANSACTION VALUE

The amount a specific buyer may pay, potentially reflecting control, synergies or financing capacity.

A market price of €50 does not prove that the business is worth exactly €50 per share.

Valuation is a defensible estimate — not the discovery of one hidden exact number.

The Question We Are Ultimately Trying to Answer

MARKET PRICE

€50 / share • Observable today

ESTIMATED INTRINSIC VALUE

€62 / share • Forecasts, cash flows & risk

IMPLIED VALUATION GAP: +24%

Reported Financials → Normalize → Forecast → FCFF

FCFF → WACC → Enterprise Value

Enterprise Value → Equity Value → Value per Share

Compare estimated value per share with the current market price.

The conclusion is conditional on the assumptions underlying the valuation.

The Valuation Landscape

INTRINSIC

Expected future cash flows → present value

- DDM
- FCFE
- FCFF

RELATIVE

Comparable-company
market pricing
Trading multiples

TRANSACTION & LBO

Buyer prices and investor
returns
Precedent transactions
LBO

ASSET BASED

Underlying assets and
strategic flexibility

- NAV
- Liquidation

No method is universally superior; method choice depends on the business, purpose and perspective.

What Exactly Are We Valuing?

OPERATING BUSINESS

FCFF → WACC → Enterprise Value

COMMON SHAREHOLDERS

FCFE → Cost of Equity → Equity Value

The cash flow, discount rate and value concept must match.

- FCFF belongs to debt and equity capital providers.
- FCFE is the residual cash flow available to common equity.
- Mixing FCFF with the cost of equity is a framework error.

Enterprise Value → Equity Value

THE CORE BRIDGE

$$\text{Equity Value} = \text{Enterprise Value} - \text{Debt} + \text{Cash} + \text{Other Non-operating Assets}$$

ENTERPRISE VALUE

Value of operating assets

– DEBT

Claims of lenders

+ CASH

Excess / non-operating cash

EQUITY VALUE

Residual to shareholders

Before comparing two valuation numbers, ask: are both enterprise values or both equity values?

Quick Exercise — Enterprise Value to Equity Value

Aster SA

- Operating business value: €1,250m
- Interest-bearing debt: €310m
- Cash: €85m
- Non-operating investments: €25m
- Shares outstanding: 40m

YOUR TASK

1. Estimate equity value.
2. Estimate value per share.
3. Why is cash added?
4. If FCFF was discounted, what value did we obtain before this bridge?

3 minutes individually → 3 minutes with your neighbour.

Financial Statements Through a Valuer's Lens

INCOME STATEMENT

What is sustainable operating profit?

Which items should be normalized?

BALANCE SHEET

How much capital is tied up in operations?

What is operating vs financing?

CASH FLOW STATEMENT

How much must the business reinvest?

What cash is available to investors?

Accounting reports the past; valuation reorganizes it around future operating cash generation.

Operating vs. Non-operating Items

OPERATING

Required to run the core business.

Examples

- Inventory
- Trade receivables
- Trade payables
- Operating PPE
- Core operating expenses

NON-OPERATING / FINANCING

Not required for core operations or primarily financing-related.

Examples

- Excess cash
- Marketable securities
- Interest-bearing debt
- Non-core investments

Classification affects both operating performance and the EV-to-equity bridge.

Reported Earnings $\neq$ Sustainable Earnings

REMOVE

Non-operating gains
and losses

ADJUST

Genuinely unusual or
non-recurring items

RETAIN

Economically recurring
costs, even if labelled
“one-off”

RECLASSIFY

Items recorded in the
wrong operating /
financing bucket

Normalization requires economic judgment — not automatic add-backs.

Worked Example — Normalize EBIT

Reported EBIT €420m

Restructuring charge €35m

Warehouse disposal gain €20m

Recurring litigation expense €8m

One-off cyber incident €12m

POSSIBLE NORMALIZED EBIT

$$420 + 35 - 20 + 12 = €447m$$

Why do we retain the recurring litigation expense?

The relevant question is economic recurrence, not management's label.

Pair Discussion — Would You Normalize It?

A

Factory closure cost after a permanent restructuring

B

Annual “restructuring” charge for the fourth consecutive year

C

Gain from selling unused land

D

Temporary raw-material spike likely to reverse

E

CEO compensation materially above peers

For each item: adjust, retain, or investigate further — and defend your reasoning.

NOPAT — Operating Profit After Operating Taxes

AFTER-TAX OPERATING PROFIT

$$\text{NOPAT} = \text{Normalized EBIT} \times (1 - \text{Operating Tax Rate})$$

- Starts from operating profit before financing costs.
- Applies taxes to the operating business, not to levered net income.
- Provides the operating-profit base for ROIC and FCFF.
- Use normalized EBIT when reported EBIT is distorted.

NOPAT isolates operating economics from financing choice.

Invested Capital — What Has Been Put Into Operations?

OPERATING APPROACH

Invested Capital = Operating Assets – Operating Liabilities

FINANCING APPROACH

≈ Debt + Equity – Non-operating Assets

- Include assets required to generate operating profit.
- Subtract non-interest-bearing operating liabilities.
- Exclude excess cash and other non-operating assets.
- Keep the NOPAT numerator and invested-capital denominator economically consistent.

ROIC is meaningful only when profit and capital are defined consistently.

ROIC — The Economic Productivity of Invested Capital

RETURN ON INVESTED CAPITAL

$$\text{ROIC} = \text{NOPAT} \div \text{Invested Capital}$$

ROIC > WACC

Potential value creation: investment earns above its opportunity cost.

ROIC $\approx$ WACC

Growth may add scale without creating much incremental economic value.

ROIC < WACC

Growth can destroy value if new investment earns inadequate returns.

Growth becomes valuable only when the economics of reinvestment are attractive.

Worked Example — NOPAT, Invested Capital & ROIC

- Normalized EBIT = €300m
- Operating tax rate = 25%
- Operating assets = €2,250m
- Operating current liabilities = €550m
- Excess cash = €180m

1

$$\text{NOPAT} = 300 \times 75\% = \text{€}225\text{m}$$

2

$$\text{Invested Capital} = 2,250 - 550 = \text{€}1,700\text{m}$$

3

$$\text{ROIC} = 225 \div 1,700 = 13.2\%$$

If WACC were 8%, what would the 13.2% ROIC tell you?

Growth Is Not Free

FUNDAMENTAL OPERATING RELATIONSHIP

$$\text{Growth} \approx \text{Reinvestment Rate} \times \text{ROIC}$$

HIGH ROIC

Less reinvestment is needed to support a given growth rate.

LOW ROIC

More reinvestment is needed to produce the same growth.

Valuation implication

We will use this logic to challenge forward growth assumptions.

Today: understand the economic link. Next session: build and defend the forecast.

Same Growth. Very Different Cash Economics.

Both firms: NOPAT = €200m and expected operating growth = 6%

FIRM HIGHROIC

ROIC = 15%

Reinvestment rate = $6\% / 15\% = 40\%$

Reinvestment = €80m

FCFF = €120m

FIRM LOWROIC

ROIC = 8%

Reinvestment rate = $6\% / 8\% = 75\%$

Reinvestment = €150m

FCFF = €50m

Identical earnings growth can require radically different amounts of cash.

What Counts as Reinvestment?

CORE OPERATING REINVESTMENT

$$\text{Reinvestment} = \text{Net CapEx} + \text{Change in Operating NWC}$$

CAPEX

Cash invested in long-lived operating assets.

– D&A

D&A is non-cash; CapEx – D&A measures net fixed-asset investment.

+ ΔNWC

Additional operating working capital tied up to support the business.

Reinvestment is the cash cost of maintaining and expanding the operating asset base.

FCFF — Cash Flow Available to All Capital Providers

FULL CASH-FLOW FORM

$$\text{FCFF} = \text{NOPAT} + \text{D\&A} - \text{CapEx} - \Delta \text{ Operating NWC}$$

ECONOMIC FORM

$$\text{FCFF} = \text{NOPAT} - \text{Reinvestment}$$

- Independent of how the company is financed.
- Discount FCFF at WACC to obtain enterprise value.

Worked Example — From EBIT to FCFF

- EBIT = €260m
- Tax rate = 24%
- D&A = €45m
- CapEx = €70m
- Increase in operating NWC = €18m

1

$$\text{NOPAT} = 260 \times 76\% = \text{€}197.6\text{m}$$

2

$$\text{Reinvestment} = (70 - 45) + 18 = \text{€}43\text{m}$$

3

$$\text{FCFF} = 197.6 - 43 = \text{€}154.6\text{m}$$

EBITDA is not FCFF: it ignores taxes, CapEx and working-capital investment.

FCFE — Cash Flow Available to Equity Holders

EQUITY CASH FLOW

$$\text{FCFE} = \text{Net Income} + \text{D\&A} - \text{CapEx} - \Delta\text{NWC} + \text{Net Borrowing}$$

WHY NET BORROWING?

Debt issuance supplies cash; debt repayment uses cash that could otherwise be available to equity.

VALUATION MATCH

FCFE → Cost of Equity → Equity Value

FCFF is before financing claims; FCFE incorporates financing flows to common equity.

FCFF or FCFE? Keep the Framework Consistent

	FCFF	FCFE
Starting point	Operating profit / NOPAT	Net income
Financing effects	Excluded	Included
Discount rate	WACC	Cost of equity
Value obtained	Enterprise value	Equity value
Best suited when	Capital structure may change	Leverage is reasonably stable

The matching rule is fundamental: cash flow ↔ discount rate ↔ value concept.

Concept Check — Where Does Each Item Belong?

INTEREST EXPENSE

Operating earnings?

EXCESS CASH

Invested capital?

INVENTORY INCREASE

Reinvestment?

DEBT REPAYMENT

FCFF?

D&A

Cash outflow?

ASSET-SALE GAIN

Normalized EBIT?

For each item, explain the classification — not only yes or no.

Exercise — Epsilon Technologies

- Reported EBIT: €510m
- One-off restructuring charge: €30m
- Operating tax rate: 25%
- D&A: €65m
- CapEx: €105m
- Increase in operating NWC: €20m
- Beginning invested capital: €2,700m
- Debt: €900m • Cash: €140m

REQUIRED

1. Normalize EBIT
2. Calculate NOPAT
3. Calculate FCFF
4. Calculate reinvestment rate
5. Estimate ROIC
6. Estimate fundamental growth
7. Identify the correct discount rate for FCFF

Explain the economic logic connecting the numbers — not only the calculations.

Integrated Exercise — Debrief

NORMALIZED EBIT

€540m

NOPAT

€405m

REINVESTMENT

€60m

FCFF

€345m

ROIC

15.0%

FUNDAMENTAL GROWTH

≈ 2.22%

The next question is sustainability: can these operating economics persist? That is Session 2.

Common Valuation Errors

1

Mixing enterprise and equity value

2

Using reported earnings without investigating unusual items

3

Including financing items in operating profit

4

Using inconsistent NOPAT and invested capital

5

Forecasting growth without considering reinvestment

6

Treating EBITDA as free cash flow

7

Discounting FCFF at the cost of equity

8

Automatically adding back every “one-off”

Most serious valuation errors are consistency errors before they are calculation errors.